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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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# ER Diagram for an Online Library System

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**Technical Presentation**

# Introduction

**What is an ER Diagram?**

- ER (Entity-Relationship) Diagram is a graphical representation of a database.
- It shows entities, attributes, and relationships between data.
- ER diagrams are used during database design.
- They help developers understand and organize data before implementation.

**What is an Online Library System?**

- An Online Library System manages:
- Books
- Members
- Librarians
- Book borrowing and returning
- Fine calculation

# Core Technical Explanation

## Entity

- A real-world object in the database.
- Examples:
  - Book
  - Member
  - Librarian

## Attribute

- Describes the properties of an entity.
- Example (Book):
  - Book_ID
  - Title
  - Author
  - Publisher

## Relationship

* Shows how entities are connected.

Example:

* Member Borrows Book
* Librarian Manages Book

## Keys

Primary Key:

* Uniquely identifies each record.
* Example: Book_ID

Foreign Key:

* Connects two related entities.
* Example: Member_ID in the Borrow table.

# Main Components of an ER Diagram

| Entity & Attribute | Relationship & Keys |
|---|---|
| **Entity** | **Relationship** |
| A real-world object in the database. | Shows how entities are connected. |
| **Examples:** | **Examples:** |
| • Book | • Member → Borrows → Book |
| • Member | • Librarian → Manages → Book |
| • Librarian | |
| **Attribute** | **Keys** |
| Describes the properties of an entity. | **Primary Key:** Uniquely identifies each record. |
| **Example (Book):** | Example: **Book_ID** |
| • Book_ID | **Foreign Key:** Connects related entities. |
| • Title | Example: **Member_ID** (in Borrow table) |
| • Author | |
| • Publisher | |

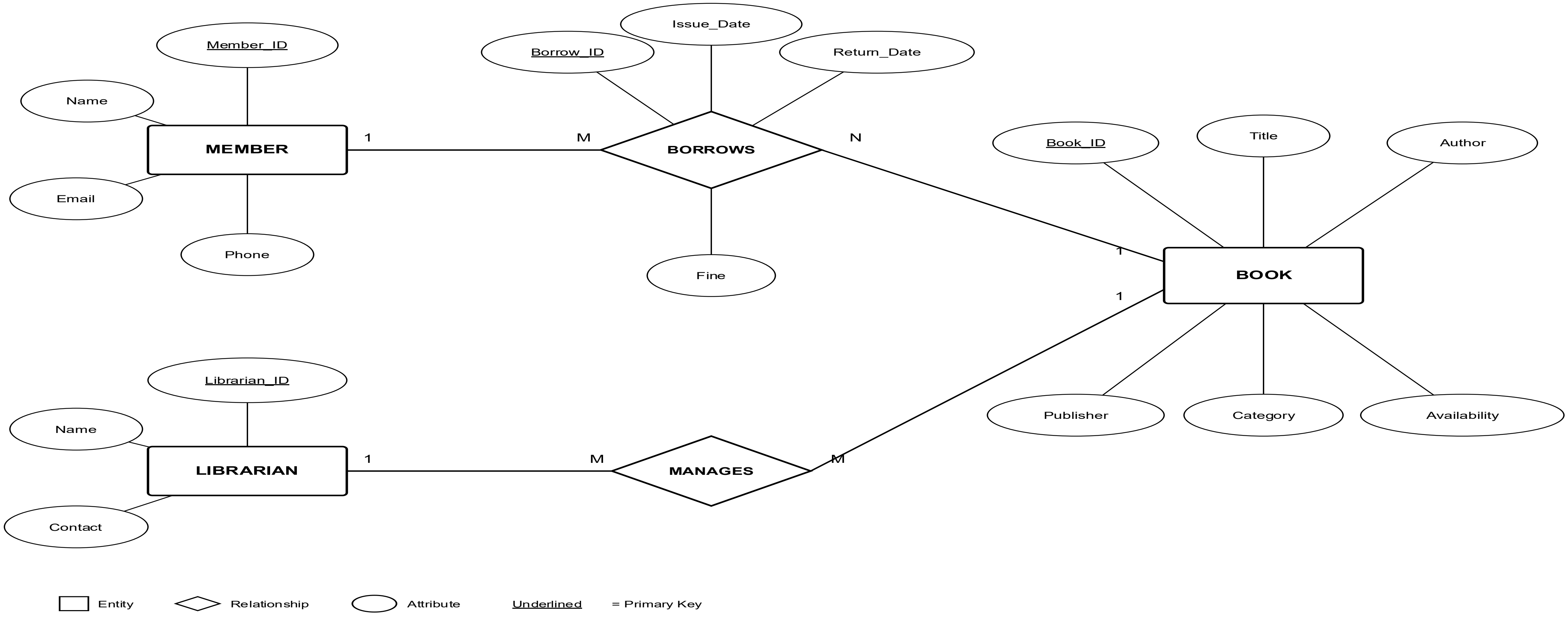
# Benefits of ER Diagram

**Advantages**

- ✓ Easy database planning
- ✓ Reduces redundancy
- ✓ Improves data consistency
- ✓ Makes database implementation easier
- ✓ Easy maintenance and modification
- ✓ Better communication between developers and users

# ER Diagram

**Online Library Management System — ER Diagram (Chen Notation)**



# Real-World Applications

**1. Online Library Management**

- Manage books, members, borrowing, and returns.

**2. Banking System**

- Store customer, account, loan, and transaction details.

**3. Hospital Management**

- Manage patients, doctors, appointments, and medical records.

**4. E-Commerce**

- Handle products, customers, orders, and payments.

**5. University Management**

- Store student, faculty, course, and examination information.

**6. Airline Reservation System**

- Manage flights, passengers, bookings, and ticket details.

# Example Use Case

- Student: Rahul
- Book: Database System Concepts
- Issue Date:
  10 July
- Return Date:
  20 July
- Fine:
  ₹20 (Late Return)

```
Student
      ↓
Search Book
      ↓
Issue Request
      ↓
Librarian Approval
      ↓
Book Issued
      ↓
Return Book
      ↓
Fine Calculation (if delayed)
```

# Conclusion and Key Takeaways

- ER Diagram is the foundation of database design.
- It organizes data into entities and relationships.
- An Online Library System efficiently manages books, members, and transactions.
- Proper ER design improves accuracy, reduces redundancy, and enhances database performance.

**Key Takeaways**

✓ Organizes data efficiently

✓ Represents relationships clearly

✓ Supports effective database development

✓ Widely used in real-world applications

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# Future Scope

**Future Enhancements**

- QR Code-based book issue and return
- RFID-enabled library management
- AI-based book recommendation
- Online book reservation
- Mobile application integration
- Cloud-based database storage

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# THANK YOU….